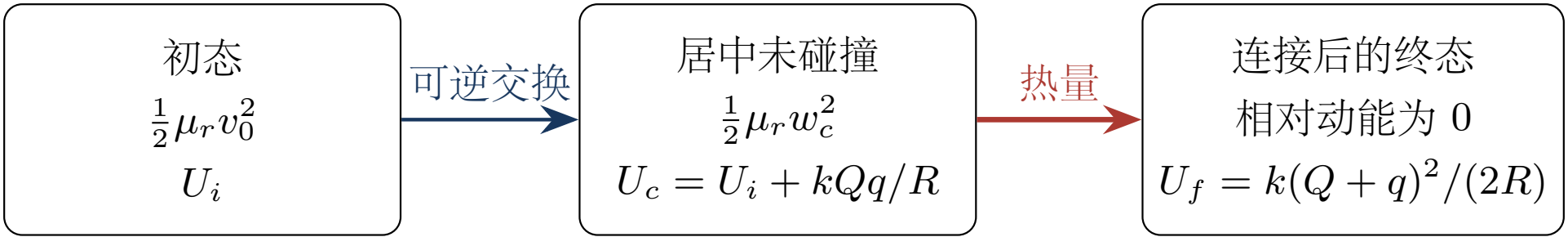

全过程质心动能保持：$\frac{1}{2}(m+M)V_c^2$

$$Q_{热} = \frac{1}{2}\mu_r w_c^2 + \frac{kq^2}{2}(1/r - 1/R)$$